

# VLSI DESIGN OPPORTUNITIES GALORE

**As the VLSI industry stands on the threshold of expansion, it faces an acute shortage of engineers. So what are you waiting for? Get, set and go!**

■ NITASHA CHAWLA

If you aspire to design the next wonder chip, it is worthwhile for you to know about very-large-scale integration (VLSI)—a stream of electronics engineering which involves putting millions and billions of transistors together logically on a tiny chip.

"This field involves packing more and more logic devices into smaller and smaller areas. Thanks to VLSI, circuits that would have taken boardful of space can now be fitted into a small space of few millimetres. This has opened avenues to do things that were not possible before," affirms Nidhi Kathuria, VLSI application engineer at EFY Tech Center—a provider of VLSI course training.

Factors such as increasing capability of an integrated circuit (IC) over the years in terms of computation power, utilisation of available area and yield have opened up new frontiers for the VLSI industry to grow. Since VLSI is a

niche industry, skillsets in this industry are short in supply and therefore in great demand. Hence there is tremendous scope and growth for those who choose VLSI design as a career.

## Scope of VLSI

According to industry experts, the Indian VLSI industry requires anywhere between 10,000 and 20,000 highly trained engineers at present. According to a joint report of India Semiconductor Association (ISA) and Ernst & Young, the semiconductor design industry in India is expected to log a compound annual growth rate of 17.3 per cent over the next three years to reach \$10.6 billion in 2012.

There are a variety of career opportunities in product companies, design services companies and electronic design automation (EDA) companies. Product and application domains of VLSI include mobile and consumer electronics, computing, telecommu-

nications and networking, data processing, automotive, healthcare and industrial applications.

"In simple words, VLSI circuits are everywhere from your computer to your car, your brand new state-of-the-art digital camera, cellphones, and whatever electronics item you have," says Kathuria.

Consumer demand for electronic products constitutes about 60 per cent of semiconductor sales today. With consumerisation comes the pressure to lower costs, retain product differentiation, manage volatile cycles and win the time-to-market race. "These pressures drive the demand for developing complex system-on-chip (SoC) devices, thereby creating a demand for VLSI skillsets," affirms Vasudevan Aaghoramoorthy, vice president, semiconductor and systems, Wipro Technologies.

"Today, there is a huge demand in the industry for VLSI designers to develop field-programmable gate array (FPGA) implementations, application-specific integrated circuit (ASIC) designs and SoCs. The VLSI industry has an added attraction of being a strong domain for patents filing, invention disclosures and applied research," he adds.

## Who should opt for it?

Being a fast-changing technology area, VLSI design is an extremely challenging and creative sector that offers exciting opportunities and fast growth for engineers. An entry-level engineer in IC design should possess a BE or B.Tech degree in electronics and telecommunications or computer science

